Laiba Irfan
31736
Operating Systems
Lab 4: Traps

(1) answers :

```
Laiba Irfan | ERP 31736 ~/xv6-labs-2026$    cat answers-traps.txt
Q1. Which registers hold function arguments? Which holds 13 in main's printf?
    a0-a7 hold the first eight arguments. For printf("%d %d\n", f(8)+1, 13):
    a0 = format string, a1 = 12, a2 = 13. So 13 is in a2.

Q2. Where are the calls to f and g in main?
    There are none. The compiler worked out f(8)+1 = 12 at compile time
    and just does "li a1,12". g is also inlined into f.

Q3. At what address is printf?
    0x746 (from user/call.asm: "0000000000000746 <printf>:")

Q4. What is in ra just after the jalr to printf in main?
    The address of the instruction right after the jalr (jalr address + 4),
    so printf's ret returns to the next line of main.

Q5. Output of: unsigned int i = 0x00646c72; printf("H%x Wo%s", 57616, (char*)&i);
    "He110 World". 57616 = 0xe110. RISC-V is little-endian, so i is stored
    as bytes 72 6c 64 00 = 'r' 'l' 'd' '\0', giving "rld".
    Big-endian: i would need to be 0x726c6400. 57616 would not change,
    because %x prints the value, which doesn't depend on byte order.

Q6. printf("x=%d y=%d", 3); what is printed after y=?
    An unpredictable number: whatever happened to be in register a2,
    because the caller never set it.Laiba Irfan | ERP 31736 ~/xv6-labs-2026$
```

(2) backtrace :

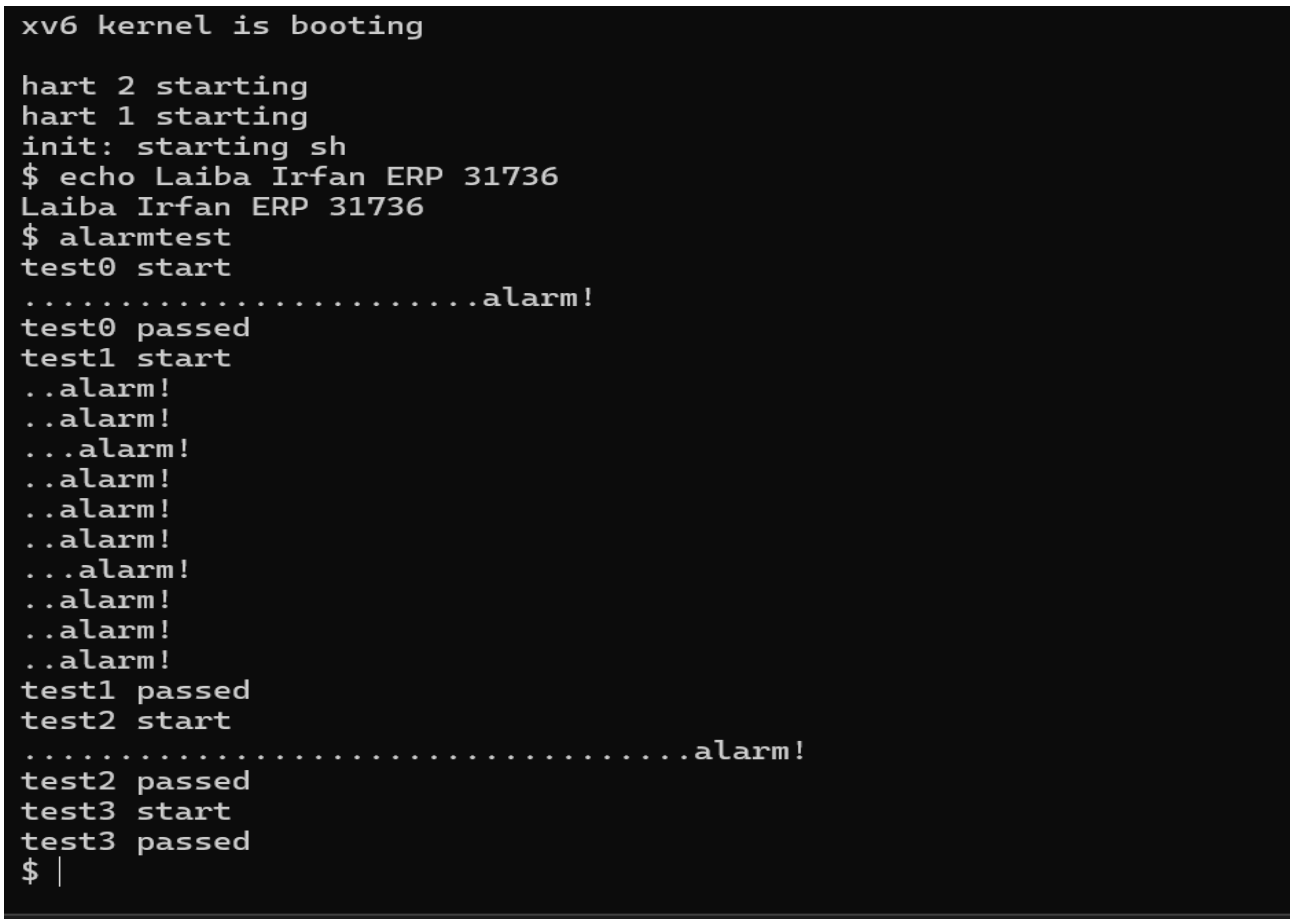
```
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$ bttest
backtrace:
0x0000000080001f16
0x0000000080001df8
0x0000000080001b40
$
```

(3) addr2line :

```
Laiba Irfan | ERP 31736 ~/xv6-labs-2026$    riscv64-linux-gnu-addr2line -e kernel/kernel
   0x0000000080001f16
   0x0000000080001df8
   0x0000000080001b40./kernel/sysproc.c:74
./kernel/syscall.c:150 (discriminator 1)

./kernel/trap.c:81
```

(4) alarmtest :


xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ echo Laiba Irfan ERP 31736
Laiba Irfan ERP 31736
$ alarmtest
test0 start
.....alarm!
test0 passed
test1 start
..alarm!
..alarm!
...alarm!
..alarm!
..alarm!
..alarm!
...alarm!
..alarm!
..alarm!
..alarm!
test1 passed
test2 start
.....alarm!
test2 passed
test3 start
test3 passed
$

(5) make-grade :

```
== Test answers-traps.txt ==
answers-traps.txt: OK
== Test backtrace test ==
$ make qemu-gdb
backtrace test: OK (5.4s)
== Test running alarmtest ==
$ make qemu-gdb
(4.0s)
== Test   alarmtest: test0 ==
  alarmtest: test0: OK
== Test   alarmtest: test1 ==
  alarmtest: test1: OK
== Test   alarmtest: test2 ==
  alarmtest: test2: OK
== Test   alarmtest: test3 ==
  alarmtest: test3: OK
== Test usertests ==
$ make qemu-gdb
usertests: OK (106.1s)
== Test time ==
time: OK
Score: 95/95
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```

(6) Challenge :

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ echo Laiba Irfan ERP 31736
Laiba Irfan ERP 31736
$ bttest
backtrace:
0x0000000080001f16 sys_pause+0xc (sysproc.c:72)
0x0000000080001df8 syscall+0x3a (syscall.c:150)
0x0000000080001b40 usertrap+0xa6 (trap.c:68)
$
```